

Domain 6 Lesson 1

Python v2

io

Programming languages come with pre-built code libraries that focus on specific functions. In Python, these libraries are called modules and must be imported before they can be used. The io module enables users to store and access data from memory, eliminating the need to work with physical files.

Purpose

Upon completing this project, you will become more familiar with the io module.

Steps for Completion

- Review the code, then answer the questions regarding the io module.

```
611-io.py > ...
1  import io
2
3  game_stream = io.StringIO()
4
5  game_stream.write("The game has started.\n")
6  game_stream.write("Here is your first question. \n")
7
8  game_stream.seek(1)
9  print(game_stream.read())
```

- What does the seek function on line 8 of this code do?
 - It finds the first character from the stream.

 - What will be the output of running this code?
 - he game has started. Here is your first question.

- What is the advantage of using the io module instead of physical files?
 - You can manage streams in memory for faster processing times and less disk usage.

 - Label the following statements as true or false.
 - False Bytes IO is a built-in class that only allows in-memory text data storage.
 - True String IO is a built-in io class that allows users to input text strings into memory and retrieve them later.

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 6

Topic: System and Command-Line Operations

Subtopic: io

Objectives covered

6 Operations Using Modules and Tools

6.1 Perform basic file system and command-line operations by using built-in modules

6.1.1 io

OS

Python's os module enables programmers to perform various operating system-based actions. The module offers versatile functions allowing users to rename files, access and manipulate environment variables, and execute shell commands from Python code. By offering the ability to manipulate files within a Python application, the os module eliminates the need to use another source to modify those files, thus simplifying working with them.

Purpose

Upon completing this project, you will better understand the os module.

Steps for Completion

1. Review the code, then answer the questions regarding the os module.

```
612-os.py > ...
1  import os
2
3  print("Your current directory is:", os.getcwd())
4
5  for text_file in os.listdir():
6      if text_file.endswith('.txt'):
7          print(text_file)
```

- a. What does line 3 do?
 - i. Print the current working directory
 - b. What does os.listdir do in line 5?
 - i. Lists the files and folders in the current working directory
2. Open the **612-os.py** file from your Domain 6 Student folder.
 3. Run the code and observe the list of files within the directory.
 4. Add code on line 9 to rename the **601-message.txt** file **OLD601-message.txt**
 5. Cut the for loop and paste it after the rename code.
 6. Run the code to ensure the 601-message.txt file is renamed.
 7. Save the file as **612-os-completed**

Project Details

Project file

612-os.py

Estimated completion time

5-10 minutes

Video reference

Domain 6

Topic: System and Command-Line Operations

Subtopic: os

Objectives covered

6 Operations Using Modules and Tools

6.1 Perform basic file system and command-line operations by using built-in modules

6.1.2 os

os.path

The os.path sub-module provides methods to check for the existence of a file or directory in a specific location. The ability to do this is beneficial when an application depends on particular directory structures and needs to ensure they exist.

Purpose

In this project, you will practice using the os.path submodule.

Steps for Completion

1. Open the **613-ospath.py** file from your Domain 6 Student folder.
2. Change line 1 to limit this import to only the os.path submodule.
3. On line 15, as part of the else block of code, write **file_path**
os.path.join('613-test' , '613-test.txt')
4. On line 16, add code that checks if the path on line 15 is a legitimate file path.
5. On line 17, add a print statement to confirm the file exists.
6. Run the code. Does the file exist? How do you know?
 - a. It does not due to the print statement
7. Save the file as **613-ospath-completed**

Project Details

Project file

613-ospath.py

Estimated completion time

5-10 minutes

Video reference

Domain 6

Topic: System and Command-Line Operations

Subtopic: os.path

Objectives covered

6 Operations Using Modules and Tools

6.1 Perform basic file system and command-line operations by using built-in modules

6.1.3 os.path

sys

The sys module provides access to some of the system-specific functionalities in Python. It allows users to import other modules, manage files, and handle command-line arguments. It also offers access to system-based commands like changing file paths and terminating an application.

Purpose

Upon completing this project, you will better understand the sys module.

Steps for Completion

1. Review the code, then answer the questions regarding the sys module.

```
614-sys.py > ...
1  import sys
2
3  game_lives = 1
4  while game_lives < 3:
5      print("Game in progress")
6      game_lives +=1
7  sys.exit()
```

- a. What does line 7 of this code do?
 - i. Exit the process

```
614-command.py > ...
1  import sys
2
3  filename = sys.argv[1]
4  print (f" {filename} has been specified.")
5
6  with open (filename, 'r') as file:
7      content = file.read()
8      print(content)
```

- b. What is sys.argv[1] on line 3?
 - i. The first command line argument
- c. Where do you run this file to ensure the sys.argv command works?
 - i. The command line

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 6

Topic: System and Command-Line Operations

Subtopic: sys

Objectives covered

6 Operations Using Modules and Tools

6.1 Perform basic file system and command-line operations by using built-in modules

6.1.4 sys (importing modules, opening, reading and writing files, command-line arguments)

Review 6.1

Purpose

Upon completing this project, you will become more familiar with identifying and importing Python libraries to code, and using methods from these built-in libraries.

Steps for Completion

1. Open the **614-analyze.py** file from your Domain 6 Student folder.
2. Complete the import statement on line 1 to import the module that works with input and output streams.
3. Complete the code on line 3 to use the method needed to enable streaming text to memory.
4. Return to the videos to compare your answers to the answers in the videos.

Project Details

Project file

614-analyze.py

Estimated completion time

5 minutes

Video reference

Domain 6

Topic: System and Command-Line Operations

Subtopic: sys; Review 6.1

Objectives covered

6 Operations using Modules and Tools

6.1 Perform basic file system and command-line operations by using built-in modules

Domain 6

Lesson 2

Python v2

Math

Python's Math module is a comprehensive computational category that provides various math functions such as fabs, ceil, floor, trunc, fmod, frexp, and nan. All these functions are readily available in Python's standard library, eliminating the need for additional installations.

Purpose

Upon completing this project, you will become more familiar with the math submodule.

Steps for Completion

- Match the math function to its description.

A. fabs	B. ceil	C. floor
D. fmod	E. frexp	

- C Returns the next integer up after a decimal number.
 - A Returns a number's absolute distance from zero.
 - B Returns an integer without a decimal number.
 - D Returns the modulus, also known as a remainder, from dividing two numbers in floating decimal format.
 - B Returns the next integer up after a decimal number.
- Which math function is similar to ceil and floor?
 - frexp??

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 6

Topic: Solve Complex Problems

Subtopic: Math

Objectives covered

6 Operations Using Modules and Tools

6.2 Solve complex computing problems by using built-in modules

6.2.1 Math (fabs, ceil, floor, trunc, fmod, frexp, nan, isnan, sqrt, isqrt, pow, pi)

isnan, sqrt, isqrt, and pi

The Math module helps solve complex math problems related to statistics, geometry, trigonometry, and others without writing much code. These math functions can help with gaming apps, as many have math or statistics-based functionality. Some additional math functions include nan, isnan, sqrt, isqrt, and pi.

Purpose

Upon completing this project, you will become more familiar with the isnan, sqrt, isqrt, and pi math functions.

Steps for Completion

1. What does the isnan function do?

a. Checks if a value is Not a Number.

2. Label the following statement as true or false.

a. true nan is a result, not a function.

3. What does isqrt return?

a. Rounds a square root to the nearest integer

4. What will this print statement return if $y = 3$ and $x = 4$?

```
print(math.pow(y,x))
```

a. 81.0

5. What will this print statement return if $x = 3$?

```
print(math.pi * math.pow(x, 2))
```

a. 28.274333882308138

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 6

Topic: Solve Complex Problems

Subtopic: isnan, sqrt, isqrt, and pi

Objectives covered

6 Operations Using Modules and Tools

6.2 Solve complex computing problems by using built-in modules

6.2.1 Math (fabs, ceil, floor, trunc, fmod, frexp, nan, isnan, sqrt, isqrt, pow, pi)

datetime

Every programming language, including Python, offers several ways to format dates and times. Some apps use fixed dates, while others require the current date, such as when a game logs the current date and time as the starting point for a game based on timing.

Purpose

In this project, you will practice working with the datetime module.

Steps for Completion

1. Open the **622-datetime.py** file from your Domain 6 Student folder.
2. On line 2, delete the comment and add code to show the current date and time.
3. On line 3, delete the comment and add code to change the current date and time format to mm-dd-yy.
4. Run the code. What is the result?
 - a. The current date and time is: 09-30-26
5. On line 3, replace the comment with the code required to show the current date and time format as mm-dd-yyyy hh:mm.
6. On line 4, add a print statement to return the number of the weekday.
7. On line 5, add a print statement to return the name of the weekday.
8. Save the file as **622-datetime-completed**

Project Details

Project file

622-datetime.py

Estimated completion time

5-10 minutes

Video reference

Domain 6

Topic: Solve Complex Problems

Subtopic: datetime

Objectives covered

6 Operations Using Modules and Tools

6.2 Solve complex computing problems by using built-in modules

6.2.2 datetime(now, strftime, weekday)

Domain 6

Lesson 3

Python v2

Random with Numbers

The random module offers multiple methods to generate random numbers or values from a given list of values. In gaming, random numbers play a significant role as many games need different events to happen each time someone plays those games to keep them fun and interesting.

Purpose

Upon completing this project, you will become more familiar with using the random module with numbers.

Steps for Completion

1. Review the code, then answer the questions regarding the random module.

```
623-random.py > ...
1  from random import randrange
2  for i in range(10):
3      print(randrange(5,20))
```

- a. What will running this code return?
 - i. 10 random numbers between 5-20
- b. What can be added to the randrange method on line 3 to return odd numbers only?
 - i. Adding 2 to the step parameter
- c. How can you make this return a list of even numbers?
 - i. Changing randrange to randint

```
623-random.py > ...
1  from random import randrange
2  for i in range(10):
3      print([randint(1,20)])
```

- d. What will happen if you run this code?
 - i. It will error because randint was not imported
- e. How can you fix this code?
 - i. Change randrange in the import to randint
2. What does the random method return?
 - a. Random float from 0-1

Project Details

Project file

N/A

Estimated completion time

10 minutes

Video reference

Domain 6

Topic: Solve Complex Problems

Subtopic: Random with Numbers

Objectives covered

6 Operations Using Modules and Tools

6.2 Solve complex computing problems by using built-in modules

6.2.3 random (randrange, randint, random, shuffle, choice, sample)

Purpose

Steps for Completion

1. Match the random module with its description.

- C Returns multiple random values from a list.
- A Rearranges the order of a list.
- B Chooses a value, at random, from a list of values.

- a. C

Project Details

Project file

N/A

Estimated completion time

5 minutes

Video reference

Domain 6

Topic: Solve Complex Problems

Subtopic: Random with Lists

Objectives covered

6 Operations Using Modules and Tools

6.2 Solve complex computing problems by using built-in modules

6.2.3 random (randrange, randint, random, shuffle, choice, sample)

Review 6.2

Purpose

Upon completing this project, you will better understand how to build a small app that calculates using random numbers.

Steps for Completion

1. Build a Python app that does the following:
 - a. Captures two whole numbers via inputs from a user
 - b. Performs a calculation of those two numbers. The calculation itself is random, with a random operator of a plus sign, minus sign, or multiplication sign.
 - c. The results should print in this format: $3 + 5 = 8$
2. Save the file as **623-final** to your Domain 6 Student folder.

Project Details

Project file

N/A

Estimated completion time

15 minutes

Video reference

Domain 6

Topic: Solve Complex Problems

Subtopic: Random with Lists; Final Review

Objectives covered

6 Operations using Modules and Tools

6.2 Solve complex problems by using built-in modules